

# Crowdfunding Platform - Detailed Requirement Analysis

The Crowdfunding Platform will serve as an online marketplace where users can post projects and seek funding from investors. The platform will allow project creators to define their campaigns, milestones, rewards, and funding goals, while investors can pledge amounts and track project progress. The platform also integrates a reward system, a payment gateway for secure transactions, and tools for feedback, comments, and reporting.

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## Entities

### 1. Project:

- Represents the individual projects posted on the platform.
- Each project has a creator, a funding goal, and progress tracked through milestones.

### 2. Investor:

- Represents users who fund projects by pledging money.
- Investors can receive rewards based on their contribution.

### 3. Campaign:

- Each project has a campaign associated with it, representing the entire lifecycle of fundraising.
- Contains start and end dates, and monitors progress towards the funding goal.

### 4. User:

- Represents individuals who can act as both project creators or investors.
- Holds personal information and account credentials.

### 5. Pledge:

- Represents the amount pledged by investors for specific projects.
- Includes the details of the investor, project, and amount pledged.

### 6. Payment:

- Handles all financial transactions between investors and projects.
- Supports payments, refunds, and disbursements when goals are met.

### 7. Reward:

- Represents the reward tiers offered to investors based on their pledged amounts.
- Each reward is linked to a project and its associated pledge levels.

### 8. Comment:

- Allows users (investors) to post comments on a project's campaign page.

### 9. Category:

- Represents the different categories of projects (e.g., technology, arts, education).
- Helps investors find projects in areas of interest.

### 10. Goal:

- Represents the financial goal that a project aims to achieve.

- Includes both the total amount and any stretch goals for additional milestones.

#### 11. Notification:

- Represents system notifications sent to users.
- Investors can be notified of project updates, goal achievements, or new comments.

#### 12. Invoice:

- Represents the receipt of each transaction or pledge made by an investor.

#### 13. Milestone:

- Tracks the project's progress at different stages.
- Creators can set specific milestones that investors can track.

#### 14. Feedback:

- Allows investors to provide feedback after the project's completion.

#### 15. Report:

- A module that generates reports for project funding, investor activity, and trends.

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### Modules

#### 1. Project Management:

- **Description:** Enables users to create, edit, and update project campaigns.
- **Features:**
  - Project creators can define project details, funding goals, categories, and milestones.
  - Track and update project status.
  - Link rewards to funding levels.

#### 2. Investment Tracking:

- **Description:** Allows investors to track their pledges and view the project funding status in real-time.
- **Features:**
  - Investors can view their contribution history and total project funding.
  - Visual progress bar to track current funding vs. goal.

#### 3. Reward System:

- **Description:** Provides reward tiers to investors based on their pledge amounts.
- **Features:**
  - Project creators can set up multiple reward levels with descriptions.
  - Investors can choose the reward tier while making a pledge.
  - Automatic updates once rewards are claimed or exhausted.

#### 4. Payment Gateway:

- **Description:** Facilitates secure payment transactions, including pledges, payments, and refunds.

- **Features:**

- Supports multiple payment methods (credit card, PayPal, bank transfer).
- Automatic refunds if funding goals are not met.
- Integration with external payment processors.

## 5. Milestone Tracking:

- **Description:** Enables project creators to define key milestones and track progress.

- **Features:**

- Display of milestone status and completion percentage to investors.
- Notifications for investors when key milestones are reached.
- Milestone reporting for project creators.

## 6. Comment and Feedback:

- **Description:** Enables communication between project creators and investors through comments and feedback.

- **Features:**

- Users can post comments on a project page.
- Investors can provide feedback on project completion.
- Project creators can respond to comments and feedback.

## 7. Reporting Module:

- **Description:** Provides analytical tools for tracking project performance, funding success, and user activity.

- **Features:**

- Generate detailed reports on funding progress, trends, and investor activity.
- Insights into project categories that perform well.
- Reports on milestone completion rates and user engagement.

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## Workflow

### 1. User Registration & Authentication:

- Users sign up and authenticate using credentials or third-party services (Google, Facebook).
- They can register either as project creators or investors (or both).

### 2. Project Creation:

- Project creators define project details such as title, description, category, funding goal, milestones, and rewards.
- Campaign start and end dates are set.
- The project is published and listed in the respective category.

### 3. Investors Discover Projects:

- Investors browse or search projects by category, popularity, or trending status.

- They view project details, funding goals, milestones, rewards, and comments.

#### 4. Pledge and Payment:

- Investors select the pledge amount and corresponding reward tier.
- Payment is processed through the payment gateway.
- Upon successful payment, an invoice is generated, and the project funding status is updated.

#### 5. Milestone Tracking:

- Project creators update milestones as they complete different phases of the project.
- Investors receive notifications for milestone completion.

#### 6. Commenting and Feedback:

- Investors leave comments on the project page during the campaign.
- After project completion, investors can provide feedback based on the outcome.

#### 7. Reward Distribution:

- Project creators distribute rewards to investors based on their pledge level after the funding goal is met.

#### 8. Reporting:

- Project creators can generate reports on funding progress, investment trends, and milestone tracking.
- Investors can view their funding history, active campaigns, and pledged projects.

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### Database Entities Overview

| Entity       | Key Attributes                                                        |
|--------------|-----------------------------------------------------------------------|
| Project      | project_id, title, description, goal_amount, start_date, end_date     |
| Investor     | investor_id, name, email, total_pledges                               |
| Campaign     | campaign_id, project_id, funding_goal, progress, start_date, end_date |
| User         | user_id, name, email, role, password                                  |
| Pledge       | pledge_id, project_id, investor_id, amount, reward_id                 |
| Payment      | payment_id, pledge_id, amount, status, method                         |
| Reward       | reward_id, project_id, description, min_pledge_amount                 |
| Comment      | comment_id, project_id, investor_id, content, timestamp               |
| Category     | category_id, name                                                     |
| Goal         | goal_id, project_id, amount, milestone_description                    |
| Notification | notification_id, user_id, message, status                             |
| Invoice      | invoice_id, payment_id, user_id, amount, date                         |

| Entity    | Key Attributes                                           |
|-----------|----------------------------------------------------------|
| Milestone | milestone_id, project_id, description, completion_status |
| Feedback  | feedback_id, investor_id, project_id, rating, comments   |
| Report    | report_id, user_id, type, generated_date, content        |

This **Crowdfunding Platform** will create a collaborative space where users can share their innovative ideas, secure funding, and build communities of supporters. The platform will streamline project creation, investment, and interaction between creators and investors.

# Crowdfunding Platform - Detailed Modules Description

Below is the comprehensive description of each module for the Crowdfunding Platform project. The platform allows users to post projects, seek funding, manage campaigns, track progress, and interact with investors.

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## 1. Project Management Module

### Description:

The Project Management module enables users (project creators) to create, edit, update, and manage campaigns for their projects. It allows defining all essential details such as project goals, milestones, rewards, and more.

### Key Features:

- **Create Project Campaigns:** Project creators can set up a project with details like title, description, category, goal amount, duration, and start/end dates.
- **Edit/Update Projects:** Allows creators to make changes to the project details after the campaign is live (within certain limits).
- **Project Categories:** Projects can be categorized into predefined categories (e.g., technology, art, education) to improve visibility and organization.
- **Set Funding Goals:** Define the target funding amount for the project. The project's progress is measured against this goal.
- **Reward System:** Project creators can define reward tiers based on different pledge amounts.
- **Project Milestones:** Creators can set specific milestones (such as development phases or release stages) to track the project's progress and update investors.
- **Media Support:** Creators can upload media files such as images and videos to better present their project.

### User Flow:

- A project creator logs in → navigates to the project creation page → fills out project details → adds rewards and milestones → publishes the campaign.
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## 2. Investment Tracking Module

### Description:

The Investment Tracking module allows investors to track their pledges and monitor the funding status of projects they support in real-time. It helps both investors and project creators understand the overall progress towards funding goals.

### Key Features:

- **Pledge Overview:** Investors can view the total amount they've pledged to various projects and get detailed information on individual pledges.
- **Funding Progress Tracker:** Visual indicators like progress bars show how close the project is to reaching its funding goal.
- **Real-Time Updates:** Investors get real-time updates as more funds are pledged and milestones are reached.
- **Goal Monitoring:** Track how close the project is to achieving its funding and stretch goals.

- **Investment History:** Allows investors to view their historical investments in various projects, including rewards earned.

#### User Flow:

- An investor logs in → navigates to their investment dashboard → views a summary of pledged projects and detailed investment history → tracks funding progress.
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### 3. Reward System Module

#### Description:

The Reward System module offers project creators the ability to set up rewards for investors based on the amount pledged. It incentivizes higher contributions by offering better rewards for larger pledges.

#### Key Features:

- **Tiered Rewards:** Project creators can define multiple reward tiers (e.g., \$10, \$50, \$100) and associate different rewards with each tier.
- **Reward Description:** Detailed descriptions of each reward, including any special perks or product offers.
- **Automatic Reward Allocation:** When an investor pledges a specific amount, they automatically become eligible for the corresponding reward.
- **Reward Availability:** Option to set limits on certain rewards (e.g., "only 50 early bird rewards available").
- **Reward Delivery:** After a successful campaign, project creators are notified of the rewards that need to be distributed to each investor.

#### User Flow:

- A project creator logs in → defines reward tiers and descriptions during project creation → investors choose their desired reward when pledging → rewards are automatically distributed after the campaign.
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### 4. Payment Gateway Module

#### Description:

The Payment Gateway module is responsible for processing financial transactions, including pledges from investors and payments to project creators. It ensures secure, reliable transactions.

#### Key Features:

- **Secure Payment Processing:** Supports multiple payment methods (credit card, PayPal, bank transfer).
- **Payment Confirmation:** Investors receive a receipt/invoice for each successful transaction.
- **Refund Management:** Automatic refunds if the project does not meet its funding goal or is canceled.
- **Disbursement to Project Creators:** Funds are released to the project creator once the campaign successfully reaches its goal.
- **Payment Status Tracking:** Both investors and project creators can track the status of payments and refunds.

#### User Flow:

- An investor selects a project → enters the pledge amount → chooses a payment method → completes the payment → receives an invoice.
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## 5. Milestone Tracking Module

### Description:

The Milestone Tracking module allows project creators to define key milestones in their project's development and keep investors updated on progress. It enhances transparency and encourages investor confidence.

### Key Features:

- **Define Milestones:** Project creators can set specific milestones that indicate major phases or achievements in the project.
- **Progress Updates:** Investors receive notifications when a milestone is reached, allowing them to track project progress in real-time.
- **Milestone Reporting:** Project creators can generate reports that show the percentage of milestones completed.
- **Progress Visualization:** Visual indicators like progress bars are used to show milestone completion.

### User Flow:

- A project creator logs in → defines milestones during project creation → updates milestone status throughout the project lifecycle → investors receive notifications of progress.
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## 6. Comment and Feedback Module

### Description:

The Comment and Feedback module facilitates communication between project creators and investors. It allows investors to leave comments and feedback on projects, fostering interaction and community building.

### Key Features:

- **Commenting System:** Investors can post comments on the project's campaign page.
- **Feedback Mechanism:** After the project is completed, investors can provide feedback on the campaign and overall experience.
- **Creator Response:** Project creators can respond to comments, engage with the community, and address any concerns or questions.
- **Feedback Rating:** Investors can rate projects after completion based on their satisfaction.
- **Comment Moderation:** Admins or project creators can moderate comments to prevent spam or inappropriate content.

### User Flow:

- An investor logs in → navigates to the project page → posts a comment → the project creator or other users can respond to the comment.
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## 7. Reporting Module

### Description:

The Reporting Module provides analytical tools and reports for project creators, investors, and platform administrators. It helps track project performance, investment trends, and funding success.

### Key Features:

- **Project Funding Reports:** Generate reports showing the total amount funded, the number of investors, and the timeline of funding progress.

- **Investment Trends:** Analyze which categories are attracting the most funding and which reward tiers are the most popular.
- **Milestone Reports:** Provides data on project milestone completion and any delays or issues.
- **Investor Activity Reports:** Displays how often investors engage with the platform, their funding patterns, and preferred project types.
- **Customizable Reports:** Users can generate customized reports based on specific parameters like date range, project type, or funding status.

**User Flow:**

- A project creator or admin logs in → accesses the reporting dashboard → selects parameters for generating a report → downloads or views the report.
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## 8. Notification Module

**Description:**

The Notification Module keeps users informed about important events, updates, and activities related to their projects, investments, or platform interactions.

**Key Features:**

- **Project Updates:** Investors receive notifications when a project reaches a milestone, achieves funding goals, or faces delays.
- **Comment Alerts:** Users are notified when someone comments on their project or replies to their comment.
- **Payment Notifications:** Investors receive payment confirmations, and project creators are notified of successful fund disbursements.
- **Goal Achievements:** Users are alerted when a project reaches its funding goal or stretch goal.
- **Campaign Reminders:** Reminders for investors about campaigns that are about to end.

**User Flow:**

- A user logs in → accesses their notification center → views updates related to projects, comments, or payments.
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This **Crowdfunding Platform** provides a comprehensive system for managing crowdfunding campaigns, engaging investors, and tracking funding success, all while offering transparency and communication tools. Each module is designed to offer robust features that enhance the user experience and streamline the management of campaigns and investments.

# Crowdfunding Platform - Detailed Workflow

The following is a step-by-step, end-to-end workflow for the Crowdfunding Platform. It describes how users (both project creators and investors) interact with the system and how the platform's modules function together.

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## 1. User Registration and Authentication Workflow

This flow allows users to create accounts and log into the platform.

- **Step 1:** User visits the platform.
  - **Step 2:** User chooses to sign up as either a **Project Creator** or an **Investor**.
    - User provides basic information: name, email, password, and selects a role.
  - **Step 3:** The platform validates the details and creates the user profile.
    - Confirmation email is sent for verification.
  - **Step 4:** Once verified, the user can log in.
  - **Step 5:** Upon successful login, the user dashboard is displayed:
    - **Creators** are directed to the **Project Management Dashboard**.
    - **Investors** are directed to the **Investment Dashboard**.
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## 2. Project Creation Workflow (Project Creator)

Project creators can start campaigns to raise funds for their projects.

- **Step 1:** Project creator logs in and navigates to the **Project Management Dashboard**.
  - **Step 2:** Creator selects "Create New Project".
  - **Step 3:** Creator fills out project details (title, description, category, goal amount, funding duration, milestones, rewards, etc.).
  - **Step 4:** Project media such as images, videos, and attachments are uploaded.
  - **Step 5:** Creator reviews and finalizes the project details.
  - **Step 6:** The project is submitted for review and approval by the platform administrators.
    - **If approved**, the project is published and listed on the platform.
    - **If rejected**, feedback is given, and the creator is asked to make revisions.
  - **Step 7:** Once the project is live, the creator can track campaign performance through the **Project Management Dashboard**.
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## 3. Pledge/Investment Workflow (Investor)

Investors browse projects and make pledges to support the campaigns they are interested in.

- **Step 1:** Investor logs in and navigates to the **Explore Projects** page.
- **Step 2:** Investor searches or browses projects based on categories, funding progress, or recommendations.

- **Step 3:** Investor views detailed project pages, including descriptions, milestones, goals, rewards, and updates.
  - **Step 4:** Investor selects a project to invest in and chooses a pledge amount.
  - **Step 5:** Investor selects the reward tier based on the pledge amount.
  - **Step 6:** Investor proceeds to checkout, where they can confirm their pledge amount, select a payment method, and apply any available discount or coupon.
  - **Step 7:** Payment is processed through the **Payment Gateway Module**.
    - **If successful**, the payment is confirmed, and the pledge is recorded.
    - **If unsuccessful**, the investor is notified and can retry.
  - **Step 8:** Investor receives a confirmation of their pledge and can track it via the **Investment Dashboard**.
    - The pledged amount is added to the project's total funding amount.
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#### 4. Project Tracking and Milestones Workflow (Project Creator & Investor)

Both project creators and investors track the project's funding progress and milestones.

- **Step 1:** Project creators can set and update milestones in the **Project Management Dashboard**.
    - Each milestone represents a significant development phase (e.g., product prototype, release, etc.).
  - **Step 2:** As the project progresses, creators update the status of each milestone (e.g., milestone completion).
  - **Step 3:** Investors are notified about milestone updates and can track the progress through the **Milestone Tracking Module**.
  - **Step 4:** The **Investment Tracking Module** shows the current funding status with real-time updates on progress bars.
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#### 5. Payment Processing Workflow (Investor & Project Creator)

The platform handles all financial transactions between investors and project creators.

- **Step 1:** Investors pledge to a project, which triggers the payment process via the **Payment Gateway Module**.
  - **Step 2:** Payment details (credit card, bank transfer, etc.) are processed through a secure payment system.
  - **Step 3:** Payment confirmation is sent to both the investor and the platform.
  - **Step 4:** Funds are held in escrow until the project meets its funding goal.
    - **If the project meets its goal**, the funds are released to the project creator.
    - **If the project fails to meet its goal**, investors are automatically refunded.
  - **Step 5:** After a successful project, the project creator receives the funds and an invoice for the transaction.
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#### 6. Reward Distribution Workflow (Project Creator & Investor)

This workflow handles the allocation and distribution of rewards based on the pledge tiers.

- **Step 1:** Project creators define reward tiers during the project creation process.
  - **Step 2:** Investors choose a reward tier when pledging to a project.
  - **Step 3:** Once the project is successfully funded, the platform automatically associates rewards with corresponding investors.
  - **Step 4:** The project creator is notified of the rewards that need to be distributed to investors.
    - Physical rewards may involve shipping, while digital rewards (like e-books or exclusive content) are delivered electronically.
  - **Step 5:** Project creators mark rewards as fulfilled in the **Reward System Module**.
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## 7. Comment and Feedback Workflow (Investor & Project Creator)

The platform allows investors to leave comments and feedback on projects.

- **Step 1:** Investors navigate to a project's page and post comments regarding the campaign, project details, or any inquiries they may have.
  - **Step 2:** Project creators can respond to comments to engage with the community, answer questions, and provide updates.
  - **Step 3:** After a project is completed, investors can provide feedback and ratings for the project.
  - **Step 4:** Both comments and feedback are moderated by the platform or the project creator to ensure constructive communication.
  - **Step 5:** Comments and feedback are publicly visible to future investors.
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## 8. Notification Workflow (Investor & Project Creator)

The platform sends notifications to users based on certain activities.

- **Step 1:** Investors and project creators receive notifications when:
    - Projects are updated (e.g., new milestones, changes to rewards).
    - A comment is posted or a reply is made to an existing comment.
    - A project reaches its funding goal.
    - Milestones are achieved.
    - Payment is successfully processed or refunded.
    - A project is close to its campaign deadline.
  - **Step 2:** Users can access their notification center to view a summary of all notifications and take action accordingly.
  - **Step 3:** Notifications are sent through multiple channels (email, in-app).
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## 9. Reporting Workflow (Admin & Project Creator)

The platform generates detailed reports for project creators and administrators.

- **Step 1:** Project creators can view funding reports, milestone tracking, and investor trends through the **Reporting Module**.
  - **Step 2:** Admins can generate platform-wide reports that include:
    - Total funds raised, successful projects, categories with the highest funding.
    - Investor activity and popular projects.
  - **Step 3:** Reports can be filtered by date range, project status, category, or funding goal.
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## 10. Milestone Completion and Final Disbursement Workflow (Project Creator & Investor)

Once a project is fully funded and milestones are completed, funds are disbursed to the project creator.

- **Step 1:** Project creators update milestone progress through the **Milestone Tracking Module**.
  - **Step 2:** The system checks whether the project has reached its final milestone.
  - **Step 3:** If the final milestone is completed, the funds held in escrow are disbursed to the project creator.
  - **Step 4:** Investors are notified of the project's successful completion, and they receive their rewards.
  - **Step 5:** A final invoice is generated for the project creator.
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This detailed workflow for the **Crowdfunding Platform** ensures a seamless process for project creators to manage campaigns, for investors to support and track projects, and for both parties to handle payments, rewards, and communication effectively.